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Enlace al repositorio:

<https://github.com/JavierPerezh/PersonalIngenieriaDeDatos/tree/main/SegundoCorte/Segundo%20Parcial%20No1%20Practico>

Parcial practico MySQL. Paso a paso de la ejecución:

1. Creación de la base de datos, activación de la base de datos, creación de las tablas e inserción de los datos de prueba:

Output			
Action Output			
#	Time	Action	Message
1	13:56:55	CREATE DATABASE hospital_db CHARACTER SET utf8mb4	1 row(s) affected
2	13:56:55	USE hospital_db	0 row(s) affected
3	13:56:55	CREATE TABLE medicos (medico_id INT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(100) NOT NULL, especialidad ...	0 row(s) affected
4	13:56:55	CREATE TABLE pacientes (paciente_id INT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(100) NOT NULL, fecha_nacimie...	0 row(s) affected
5	13:56:55	CREATE TABLE consultas (consulta_id INT AUTO_INCREMENT PRIMARY KEY, medico_id INT NOT NULL, paciente_id INT NOT N...	0 row(s) affected
6	13:58:54	INSERT INTO medicos VALUES (1,"Dra. Laura Rios","Cardiologia",8500000.00,"2018-03-10",1), (2,"Dr. Carlos Mesa","Neurologia",9200000.00,"2016-...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
7	13:58:54	INSERT INTO pacientes VALUES (1,"Juan Perez","1985-04-12","Bogota","juan@mail.com","Sura"), (2,"Ana Gomez","1992-08-25","Medellin","ana@...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
8	13:58:54	INSERT INTO consultas VALUES (1,1,1,"2024-01-10","Hipertension leve",150000,"realizada"), (2,1,3,"2024-01-22","Control cardiaco",150000,"realiza...	16 row(s) affected Records: 16 Duplicates: 0 Warnings: 0

Tablas creadas:

consultas:

	Field	Type	Null	Key	Default	Extra
►	consulta_id	int(11)	NO	PRI	NULL	auto_increment
	medico_id	int(11)	NO	MUL	NULL	
	paciente_id	int(11)	NO	MUL	NULL	
	fecha_consulta	date	NO		NULL	
	diagnostico	varchar(200)	YES		NULL	
	costo	decimal(10,2)	NO		NULL	
	estado	varchar(20)	YES		programada	

Médicos:

	Field	Type	Null	Key	Default	Extra
►	medico_id	int(11)	NO	PRI	NULL	auto_increment
	nombre	varchar(100)	NO		NULL	
	especialidad	varchar(80)	NO		NULL	
	salario	decimal(12,2)	NO		NULL	
	fecha_ingreso	date	NO		NULL	
	activo	tinyint(1)	YES		1	

Pacientes:

	Field	Type	Null	Key	Default	Extra
▶	paciente_id	int(11)	NO	PRI	NULL	auto_increment
	nombre	varchar(100)	NO		NULL	
	fecha_nacimiento	date	NO		NULL	
	ciudad	varchar(60)	YES		NULL	
	email	varchar(100)	YES	UNI	NULL	
	eps	varchar(80)	YES		NULL	

Con sus valores respectivamente (consultas, médicos, pacientes):

	consulta_id	medico_id	paciente_id	fecha_consulta	diagnostico	costo	estado
▶	1	1	1	2024-01-10	Hipertension leve	150000.00	realizada
	2	1	3	2024-01-22	Control cardiaco	150000.00	realizada
	3	2	2	2024-02-05	Cefalea cronica	200000.00	realizada
	4	2	5	2024-02-18	Migraña	200000.00	cancelada
	5	3	4	2024-03-01	Control crecimiento	90000.00	realizada
	6	3	7	2024-03-14	Fiebre alta	90000.00	realizada
	7	4	6	2024-04-02	Fractura muñeca	250000.00	realizada
	8	4	1	2024-04-15	Dolor rodilla	250000.00	programada
	9	5	8	2024-05-03	Arritmia	180000.00	realizada
	10	5	2	2024-05-20	Ecocardiograma	180000.00	realizada
	11	6	3	2024-05-28	Dermatitis	120000.00	cancelada
	12	7	5	2024-06-10	Epilepsia control	220000.00	realizada
	13	7	6	2024-06-22	Resonancia	220000.00	programada
	14	8	4	2024-07-01	Vacunacion	60000.00	realizada
	15	1	8	2024-07-15	Hipertension severa	180000.00	realizada
	16	3	2	2024-07-28	Seguimiento	90000.00	programada
★	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	medico_id	nombre	especialidad	salario	fecha_ingreso	activo
▶	1	Dra. Laura Rios	Cardiologia	8500000.00	2018-03-10	1
	2	Dr. Carlos Mesa	Neurologia	9200000.00	2016-07-22	1
	3	Dra. Sofia Vega	Pediatrica	7800000.00	2020-01-15	1
	4	Dr. Andres Gil	Ortopedia	8100000.00	2019-06-01	1
	5	Dra. Paula Mora	Cardiologia	8700000.00	2017-09-30	1
	6	Dr. Ivan Cruz	Dermatologia	7500000.00	2021-04-12	0
	7	Dra. Marta Leon	Neurologia	9500000.00	2015-11-05	1
	8	Dr. Felipe Ossa	Pediatrica	7600000.00	2022-02-28	1
★	NULL	NULL	NULL	NULL	NULL	NULL

	paciente_id	nombre	fecha_nacimiento	ciudad	email	eps
▶	1	Juan Perez	1985-04-12	Bogota	juan@mail.com	Sura
	2	Ana Gomez	1992-08-25	Medellin	ana@mail.com	Compensar
	3	Luis Vargas	1978-12-03	Cali	luis@mail.com	Sura
	4	Maria Diaz	2001-06-17	Bogota	maria@mail.com	Famisanar
	5	Carlos Ruiz	1965-01-30	Barranquilla	carlos@mail.com	Compensar
	6	Lucia Herrera	1990-09-08	Bogota	lucia@mail.com	Sura
	7	Pedro Soto	2005-03-22	Cali	pedro@mail.com	Famisanar
	8	Valeria Torres	1998-11-14	Medellin	valeria@mail.com	Compensar
★	NULL	NULL	NULL	NULL	NULL	NULL

4. Consulta con joins y el resultado:

19 14:06:32 select m.nombre, p.nombre as nombre_paciente, c.fecha_consulta, c.costo from consultas c join pacientes p on c.paciente_id = p.paciente_id j... 4 row(s) returned

	nombre	nombre_paciente	fecha_consulta	costo
►	Dra. Sofia Vega	Ana Gomez	2024-07-28	90000.00
	Dra. Sofia Vega	Maria Diaz	2024-03-01	90000.00
	Dra. Sofia Vega	Carlos Ruiz	2026-04-20	90000.00
	Dra. Sofia Vega	Pedro Soto	2024-03-14	90000.00

Script completo del parcial:

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-- =====

-- CASO: Sistema de Gestion Hospitalaria

-- Tablas: medicos, pacientes, consultas

-- =====

DROP DATABASE IF EXISTS hospital_db;

CREATE DATABASE hospital_db CHARACTER SET utf8mb4;

USE hospital_db;

CREATE TABLE medicos (

    medico_id    INT AUTO_INCREMENT PRIMARY KEY,

    nombre       VARCHAR(100) NOT NULL,

    especialidad  VARCHAR(80) NOT NULL,

    salario      DECIMAL(12,2) NOT NULL CHECK (salario > 0),

    fecha_ingreso DATE NOT NULL,

    activo       TINYINT(1) DEFAULT 1

);

CREATE TABLE pacientes (

    paciente_id  INT AUTO_INCREMENT PRIMARY KEY,

    nombre       VARCHAR(100) NOT NULL,

    fecha_nacimiento DATE NOT NULL,

    ciudad       VARCHAR(60),

    email        VARCHAR(100) UNIQUE,

    eps          VARCHAR(80)

);

CREATE TABLE consultas (

    consulta_id  INT AUTO_INCREMENT PRIMARY KEY,
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medico_id    INT NOT NULL,

paciente_id  INT NOT NULL,

fecha_consulta DATE NOT NULL,

diagnostico  VARCHAR(200),

costo        DECIMAL(10,2) NOT NULL CHECK (costo > 0),

estado       VARCHAR(20) DEFAULT "programada"

            CHECK (estado IN ("programada","realizada","cancelada")),

FOREIGN KEY (medico_id) REFERENCES medicos(medico_id),

FOREIGN KEY (paciente_id) REFERENCES pacientes(paciente_id)

);

-- =====

-- DML: DATOS DE PRUEBA

-- =====

INSERT INTO medicos VALUES

(1,"Dra. Laura Rios","Cardiologia",8500000.00,"2018-03-10",1),

(2,"Dr. Carlos Mesa","Neurologia",9200000.00,"2016-07-22",1),

(3,"Dra. Sofia Vega","Pediatria",7800000.00,"2020-01-15",1),

(4,"Dr. Andres Gil","Ortopedia",8100000.00,"2019-06-01",1),

(5,"Dra. Paula Mora","Cardiologia",8700000.00,"2017-09-30",1),

(6,"Dr. Ivan Cruz","Dermatología",7500000.00,"2021-04-12",0),

(7,"Dra. Marta Leon","Neurología",9500000.00,"2015-11-05",1),

(8,"Dr. Felipe Ossa","Pediatria",7600000.00,"2022-02-28",1);

INSERT INTO pacientes VALUES

(1,"Juan Perez","1985-04-12","Bogota","juan@mail.com","Sura"),

(2,"Ana Gomez","1992-08-25","Medellin","ana@mail.com","Compensar"),

(3,"Luis Vargas","1978-12-03","Cali","luis@mail.com","Sura"),

(4,"Maria Diaz","2001-06-17","Bogota","maria@mail.com","Famisanar"),

(5,"Carlos Ruiz","1965-01-30","Barranquilla","carlos@mail.com","Compensar"),

(6,"Lucia Herrera","1990-09-08","Bogota","lucia@mail.com","Sura"),

(7,"Pedro Soto","2005-03-22","Cali","pedro@mail.com","Famisanar"),

(8,"Valeria Torres","1998-11-14","Medellin","valeria@mail.com","Compensar");

INSERT INTO consultas VALUES

(1,1,1,"2024-01-10","Hipertension leve",150000,"realizada"),

(2,1,3,"2024-01-22","Control cardiaco",150000,"realizada"),

```

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(3,2,2,"2024-02-05","Cefalea cronica",200000,"realizada"),
(4,2,5,"2024-02-18","Migraña",200000,"cancelada"),
(5,3,4,"2024-03-01","Control crecimiento",90000,"realizada"),
(6,3,7,"2024-03-14","Fiebre alta",90000,"realizada"),
(7,4,6,"2024-04-02","Fractura muñeca",250000,"realizada"),
(8,4,1,"2024-04-15","Dolor rodilla",250000,"programada"),
(9,5,8,"2024-05-03","Arritmia",180000,"realizada"),
(10,5,2,"2024-05-20","Ecocardiograma",180000,"realizada"),
(11,6,3,"2024-05-28","Dermatitis",120000,"cancelada"),
(12,7,5,"2024-06-10","Epilepsia control",220000,"realizada"),
(13,7,6,"2024-06-22","Resonancia",220000,"programada"),
(14,8,4,"2024-07-01","Vacunacion",60000,"realizada"),
(15,1,8,"2024-07-15","Hipertension severa",180000,"realizada"),
(16,3,2,"2024-07-28","Seguimiento",90000,"programada");

describe consultas;

describe medicos;

describe pacientes;

select * from medicos;

select * from consultas;

select * from pacientes;

-- Ejercicio 4 del preparcial

-- Parte a)

insert into consultas (medico_id, paciente_id, fecha_consulta, diagnostico, costo, estado)

values (3,5,curdate(), 'Control pediatrico',90000, 'programada');

-- Parte b)

update consultas set estado = 'realizada' where consulta_id = last_insert_id();

-- Parte c)

select

    m.nombre,

    p.nombre as nombre_paciente,

    c.fecha_consulta,

    c.costo

from consultas c

join pacientes p on c.paciente_id = p.paciente_id

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```
join medicos m on c.medico_id = m.medico_id
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```
where m.nombre = 'Dra. Sofía Vega';
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